Evaluating Limits

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Direct substitution and algebraic techniques**

Evaluate each limit:

a $\lim_{x\to 1}(3x^2-2x+5) = 6.$

b $\lim_{x\to 2}\frac{x^2-4}{x-2} = \lim_{x\to 2}(x+2) = 4.$

c $\lim_{x\to 0}\frac{\sqrt{x+4}-2}{x}$ Rationalize: $\frac{(x+4)-4}{x(\sqrt{x+4}+2)} = \frac{1}{\sqrt{x+4}+2} \to \frac{1}{4}.$

(a) $= 6.$; (b) $= \lim_{x\to 2}(x+2) = 4.$; (c) Rationalize: $\frac{(x+4)-4}{x(\sqrt{x+4}+2)} = \frac{1}{\sqrt{x+4}+2} \to \frac{1}{4}.$

2 **[Foundational]** • **Target: Trigonometric and special limits**

Using $\lim_{x\to 0}\frac{\sin x}{x} = 1$, evaluate:

a $\lim_{x\to 0}\frac{\sin 5x}{x} = 5\lim_{x\to 0}\frac{\sin 5x}{5x} = 5.$

b $\lim_{x\to 0}\frac{1-\cos x}{x} = 0$ (known result: this limit is 0, not ∞).

c $\lim_{x\to 0}\frac{\sin 3x}{\sin 5x} = \frac{3}{5}\cdot\lim\frac{\sin 3x}{3x}\cdot\frac{5x}{\sin 5x} = \frac{3}{5}.$

(a) $= 5\lim_{x\to 0}\frac{\sin 5x}{5x} = 5.$; (b) $= 0$ (known result: this limit is 0, not ∞).; (c) $= \frac{3}{5}\cdot\lim\frac{\sin 3x}{3x}\cdot\frac{5x}{\sin 5x} = \frac{3}{5}.$

3 **[Foundational]** • **Target: Limits that fail to exist and the squeeze theorem**

Explain why $\lim_{x\to 0}\sin\left(\frac{1}{x}\right)$ does not exist.

As $x \to 0$, $\frac{1}{x} \to \pm\infty$, so $\sin(1/x)$ oscillates between -1 and 1 infinitely often near $x = 0$ rather than approaching a single value.

4 **[Foundational]** • **Target: Limits that fail to exist and the squeeze theorem**

Use the squeeze theorem to evaluate $\lim_{x\to 0}x^2\cos\left(\frac{1}{x}\right)$.

Since $-x^2 \le x^2\cos(1/x) \le x^2$ and $\lim_{x\to 0}(\pm x^2) = 0$, the squeeze theorem gives a limit of 0.

5 **[Foundational]** • **Target: Limits with absolute value and piecewise functions**

For $f(x) = 2x+1$ if $x < 3$, and $f(x) = x^2 - 2$ if $x \ge 3$, evaluate $\lim_{x\to 3^-}f(x)$ and $\lim_{x\to 3^+}f(x)$, and state whether $\lim_{x\to 3}f(x)$ exists.

$\lim_{x\to 3^-}f(x) = 2(3)+1 = 7$. $\lim_{x\to 3^+}f(x) = 3^2 - 2 = 7$. Since both one-sided limits equal 7, $\lim_{x\to 3}f(x) = 7$ exists.

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6 [Foundational] • Target: More algebraic limit techniques

Evaluate $\lim_{x\to 2} \frac{x^3-8}{x-2}$ by factoring with the difference-of-cubes formula.

$x^3-8=(x-2)(x^2+2x+4)$, so the limit is $\lim_{x\to 2}(x^2+2x+4)=4+4+4=12$.

7 [Foundational] • Target: Visualizing a limit numerically and graphically

The function $f(x)=\frac{\sin x}{x}$ (undefined at $x=0$) is graphed below. Use the graph to estimate $\lim_{x\to 0} f(x)$.

The graph approaches height 1 from both sides as $x\to 0$, even though the function has a hole exactly at $x=0$: $\lim_{x\to 0}\frac{\sin x}{x}=1$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Limits at infinity with different degree relationships

Evaluate each limit at infinity, based on the relative degrees of numerator and denominator:

a $\lim_{x\to\infty}\frac{4x+3}{x^2-1}$ (lower degree numerator) $=0$ (denominator grows faster).

b $\lim_{x\to\infty}\frac{x^3-1}{2x^2+5}$ (higher degree numerator) $=\infty$ (numerator grows faster; the limit does not exist as a finite number).

(a) $=0$ (denominator grows faster).; (b) $=\infty$ (numerator grows faster; the limit does not exist as a finite number).

9 [Intermediate] • Target: Limits requiring conjugate multiplication

Evaluate $\lim_{x\to 9}\frac{x-9}{\sqrt{x}-3}$ by rationalizing the denominator.

Multiply by $\frac{\sqrt{x}+3}{\sqrt{x}+3}$: $\frac{(x-9)(\sqrt{x}+3)}{x-9}=\sqrt{x}+3\to\sqrt{9}+3=6$.

10 [Intermediate] • Target: Limits requiring conjugate multiplication

Evaluate $\lim_{x\to\infty}\left(\sqrt{x^2+3x}-x\right)$ by rationalizing.

Multiply by $\frac{\sqrt{x^2+3x}+x}{\sqrt{x^2+3x}+x}$: $\frac{3x}{\sqrt{x^2+3x}+x}$. Dividing top and bottom by x: $\frac{3}{\sqrt{1+3/x}+1}\to\frac{3}{1+1}=\frac{3}{2}$.

11 [Intermediate] • Target: Piecewise limits requiring a parameter

For $f(x)=kx+1$ if $x\le 2$, and $f(x)=x^2-3$ if $x>2$, find k so that $\lim_{x\to 2} f(x)$ exists.

Need $\lim_{x\to 2^-} f(x)=\lim_{x\to 2^+} f(x)$: $2k+1=4-3=1$, so $k=0$.

12 [Intermediate] • Target: Limits involving factorable higher-degree polynomials

Evaluate $\lim_{x\to 1}\frac{x^4-1}{x-1}$ by factoring the numerator as a difference of squares twice.

$x^4-1=(x^2-1)(x^2+1)=(x-1)(x+1)(x^2+1)$, so the limit is $\lim_{x\to 1}(x+1)(x^2+1)=(2)(2)=4$.

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13 **[Intermediate]** • **Target: Limits involving factorable higher-degree polynomials**

Evaluate $\lim_{x\to -2} \frac{x^3+8}{x+2}$ using the sum-of-cubes factoring pattern.

$x^3+8=(x+2)(x^2-2x+4)$, so the limit is $\lim_{x\to -2}(x^2-2x+4)=4+4+4=12$.

14 **[Intermediate]** • **Target: Limits with absolute value and piecewise functions**

Level 2 · MediumEvaluate the trigonometric limit: $\lim_{x\to 0} \frac{1-\cos(3x)}{x^2}$.

Multiply numerator and denominator by the conjugate $1+\cos(3x)$:

$\lim_{x\to 0} \frac{(1-\cos(3x))(1+\cos(3x))}{x^2(1+\cos(3x))} = \lim_{x\to 0} \frac{1-\cos^2(3x)}{x^2(1+\cos(3x))} = \lim_{x\to 0} \frac{\sin^2(3x)}{x^2(1+\cos(3x))}$. Re-group terms to leverage $\lim_{u\to 0} \frac{\sin u}{u}=1$:
$= \lim_{x\to 0} \left[\left(\frac{\sin(3x)}{3x}\right)^2 \cdot 3^2 \cdot \frac{1}{1+\cos(3x)}\right]$. Evaluating individual limits.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: More algebraic limit techniques** • **Cross-Lesson: 1.01 Precalculus Review for Calculus**

Level 3 · AdvancedEvaluate the limit analytically without using L'Hôpital's Rule: $\lim_{x\to 0} \frac{\sqrt{1+\tan x}-\sqrt{1+\sin x}}{x^3}$.

Step 1 · Rationalize the Numerator: Multiply by $\sqrt{1+\tan x}+\sqrt{1+\sin x}$:

$\lim_{x\to 0} \frac{(1+\tan x)-(1+\sin x)}{x^3(\sqrt{1+\tan x}+\sqrt{1+\sin x})} = \lim_{x\to 0} \frac{\tan x-\sin x}{x^3(\sqrt{1+\tan x}+\sqrt{1+\sin x})}$. Step 2 · Factor Trigonometric Terms:
$\tan x-\sin x=\frac{\sin x}{\cos x}-\sin x=\sin x\left(\frac{1-\cos x}{\cos x}\right)$. Multiply numerator and denominator by $1+\cos x$.

16 **[Challenging]** • **Target: Visualizing a limit numerically and graphically** • **Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves**

Level 1 · FoundationsEvaluate the one-sided limit $\lim_{x\to 2^-} \frac{|x-2|}{x^2-4}$.

When $x\to 2^-$, \(x

17 **[Challenging]** • **Target: Limits at infinity with different degree relationships** • **Cross-Lesson: 11.01 Sequences and Series**

Level 3 · AdvancedWater flows into an initially empty tank at a rate $R(t)=120-4t$ liters/hour and leaks out at a rate $L(t)=3t$ liters/hour for $0\le t\le 24$ hours. Find the maximum volume of water in the tank and the time at which the tank becomes completely empty.

Step 1 · Volume Accumulation Function: Net rate of change is
$V'(t)=R(t)-L(t)=(120-4t)-3t=120-7t$. Since the tank is initially empty, $V(0)=0$:
$V(t)=\int_0^t (120-7u)\,du=120t-\frac{7}{2}t^2$. Step 2 · Maximum Volume: Maximum occurs when
$V'(t)=0\Rightarrow 120-7t=0\Rightarrow t=\frac{120}{7}\approx 17.14$ hours. Since x for $t>0$:

$120t-\frac{7}{2}t^2=0\Rightarrow t(120-\frac{7}{2}t)=0\Rightarrow t=\frac{240}{7}\approx 34.29$ hours. Since t.

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18 [Challenging] • Target: Limits requiring conjugate multiplication • Cross-Lesson: 12.01 Convergence Tests

Level 3 · Advanced Evaluate the improper integral or establish its divergence: $\int_0^\infty xe^{-x/2}\,dx$

Express as a limit of a definite integral: $\int_0^\infty xe^{-x/2}\,dx = \lim_{t\to\infty}\int_0^t xe^{-x/2}\,dx$ Integrate by parts on $[0,t]$: Let $u = x \Rightarrow du = dx$, and $dv = e^{-x/2}dx \Rightarrow v = -2e^{-x/2}$. $\int_0^t xe^{-x/2}\,dx = [-2xe^{-x/2}]_0^t - \int_0^t(-2e^{-x/2})\,dx = -2te^{-t/2} + 0 + 2[-2e^{-x/2}]_0^t$ $= -2te^{-t/2} - 4(e^{-t/2} - 1) = 4 - e^{-t/2}(2t+4)$ Take the limit as $t\to\infty$: By L'Hôpital's Rule.

19 [Challenging] • Target: Piecewise limits requiring a parameter • Cross-Lesson: 12.02 Power Series and Taylor Series

Level 2 · MediumFind the minimum number of subintervals n required to guarantee that the Trapezoidal approximation of $\int_1^3 \frac{1}{x}\,dx$ has an absolute error less than 10^{-4}.

Compute second derivative: $f(x) = x^{-1} \Rightarrow f'(x) = -x^{-2} \Rightarrow f''(x) = 2x^{-3} = \frac{2}{x^3}$. On $[1,3]$, $|f''(x)| = \frac{2}{x^3}$ is decreasing, so its maximum occurs at $x = 1$: $K = \frac{2}{1^3} = 2$. Using the Trapezoidal error bound with $a = 1, b = 3$: $|E_T| \le \frac{K(b-a)^3}{12n^2} = \frac{2(3-1)^3}{12n^2} = \frac{2(8)}{12n^2} = \frac{16}{12n^2} = \frac{4}{3n^2}$. Require.

20 [Challenging] • Target: Limits involving factorable higher-degree polynomials • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Level 2 · Medium Evaluate the indefinite integral using partial fraction decomposition: $\int \frac{5x-3}{x^2-2x-3}\,dx$

Factor the denominator: $x^2 - 2x - 3 = (x-3)(x+1)$. Set up partial fractions: $\frac{5x-3}{(x-3)(x+1)} = \frac{A}{x-3} + \frac{B}{x+1} \Rightarrow 5x - 3 = A(x+1) + B(x-3)$ Using Heaviside cover-up: Set $x = 3$: $5(3) - 3 = A(3+1) \Rightarrow 12 = 4A \Rightarrow A = 3$. Set $x = -1$: $5(-1) - 3 = B(-1-3) \Rightarrow -8 = -4B \Rightarrow B = 2$. Integrate term-by-term.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Direct substitution and algebraic techniques

Consider the function $f(x)$ whose graph is given below. The function is defined piecewise. For $x \le 2$, $f(x) = ax + b$. For $x > 2$, $f(x) = cx + d$. The graph of $f(x)$ passes through the point $(0, 3)$. The limit of $f(x)$ as x approaches 2 exists. Find the value of a.

a. -1 ✓

b. 1

c. 2

d. -2

For the limit of $f(x)$ as x approaches 2 to exist, the left-hand limit must equal the right-hand limit. That is, $\lim_{x\to 2^-} f(x) = \lim_{x\to 2^+} f(x)$. For $x \le 2$, $f(x) = ax + b$. So, the left-hand limit is $\lim_{x\to 2^-}(ax + b) = 2a + b$. For $x > 2$, $f(x) = cx + d$. So, the right-hand limit is $\lim_{x\to 2^+}(cx + d) = 2c + d$. Therefore, $2a + b = 2c + d$. From the graph, we can see that the function passes through $(0, 3)$. This means $f(0) = 3$. Since $0 \le 2$, we u.

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22 [Foundational] • Target: Trigonometric and special limits

Direct comparison test: if $0 \le f(x) \le g(x)$ for $x \ge a$ and $\int_a^\infty g(x)\,dx$ converges, then $\int_a^\infty f(x)\,dx$:

a. **also converges** ✓

b. must diverge

c. converges only if $f = g$

d. cannot be determined

Bounding f above by a convergent g forces f's improper integral to converge too — used to establish convergence before evaluating via FTC.

23 [Foundational] • Target: Limits that fail to exist and the squeeze theorem

$\int \frac{dx}{\sqrt{4-x^2}}$ evaluated at $x = 2\sin\theta$: after substitution:

a. $\arcsin(x/2) + C$ ✓

b. $2\cos\theta d\theta + C$

c. $\sin\theta + C$

d. $d\theta + C$

Applying Techniques of Integration principles, Result is $\arcsin(x/2) + C$. Therefore, the verified correct option is $\arcsin(x/2) + C$.

24 [Foundational] • Target: Limits with absolute value and piecewise functions

$\int_0^1 \frac{1-x^n}{1-x}\,dx = \sum_{k=1}^{n} \frac{1}{k} = H_n$. Evaluating by writing $\frac{1-x^n}{1-x} =$

a. $(1-x)^{n-1}$

b. $n/(1-x)$

c. $\ln(1-x)$

d. $1 + x + \cdots + x^{n-1}$ ✓

Applying Techniques of Integration principles, Geometric sum. Therefore, the verified correct option is $1 + x + \cdots + x^{n-1}$.

25 [Foundational] • Target: More algebraic limit techniques

Continuity of piecewise: $f(x) = \{kx, x < 2; x^2 - 2, x \ge 2$ is continuous iff $k =$

a. 1 ✓

b. 2

c. 0

d. 3

Applying Limits and Continuity principles, $2k = 4 - 2 = 2 \Rightarrow k = 1$. Therefore, the verified correct option is 1.

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26 [Foundational] • Target: Visualizing a limit numerically and graphically

Intermediate Value Theorem: If f is continuous on $[1, 4]$, $f(1) = -2$, $f(4) = 5$, must $f(c) = 0$ for some $c \in (1, 4)$?

a. No, not guaranteed

b. **Yes, by IVT ✓**

c. Only if f is differentiable

d. Only at $c = 2.5$

IVT: continuous function on $[a, b]$ takes every value between $f(a)$ and $f(b)$. Since $-2 < 0 < 5$, yes.

27 [Foundational] • Target: Limits at infinity with different degree relationships

Evaluate the one-sided limit $\lim_{x \to 3^-} \frac{|x-3|}{x-3}$.

a. 0

b. ∞

c. **-1 ✓**

d. 1

For $x \to 3^-$, $x - 3$ is negative, so $\frac{|x-3|}{x-3} = -1$.

28 [Foundational] • Target: Limits requiring conjugate multiplication

For $\int \frac{x^2+1}{x(x-1)(x+1)}\,dx$ the degree of numerator vs denominator:

a. improper; use long division

b. **proper; partial fractions directly ✓**

c. degree equal; long division

d. none

Numerator deg 2, denom deg 3: proper, use partial fractions.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Piecewise limits requiring a parameter

Trig substitution for $\sqrt{a^2 - x^2}$: use $x =$

a. $a\sin\theta$ ✓

b. $a\tan\theta$

c. $a\sec\theta$

d. $a\cos\theta$

$x = a\sin\theta \Rightarrow \sqrt{a^2 - x^2} = a\cos\theta$.

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30 [Intermediate] • Target: Limits involving factorable higher-degree polynomials

Numerical integration: Simpson's rule uses polynomials of degree:

a. 1

b. 3

c. **2 ✓**

d. 4

Parabolas (degree 2) for each pair of subintervals.

31 [Intermediate] • Target: Direct substitution and algebraic techniques

FTC in higher dimensions: $\int_a^b ablaf \cdot d\mathbf{r} = f(b) - f(a)$. This is valid for:

a. any vector field

b. **gradient fields (conservative) only ✓**

c. curl-free fields on simply-connected domains

d. divergence-free fields

FTC for line integrals: valid for gradient fields.

32 [Intermediate] • Target: Trigonometric and special limits

The substitution $u = g(x)$ in $\int_a^b f(g(x))g'(x)\,dx$ requires g to be:

a. continuous only

b. monotone only

c. C^1 with $g' \geq 0$

d. **absolutely continuous ✓**

Substitution theorem holds for abs. cont. g.

33 [Intermediate] • Target: Limits that fail to exist and the squeeze theorem

Trig substitution for $\sqrt{a^2 - x^2}$: set $x =$

a. $a\tan\theta$

b. $a\sin\theta$ **✓**

c. $a\sec\theta$

d. $a\cos\theta$

$x = a\sin\theta$ eliminates the square root.

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34 [Intermediate] • Target: Limits with absolute value and piecewise functions

Stokes' theorem: $\iint_S (abla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r}$. Relates:

a. divergence to flux

b. gradient to work

c. **curl to circulation ✓**

d. Laplacian to area

Stokes: curl flux = circulation.

35 [Intermediate] • Target: More algebraic limit techniques

Consider the function $f(x)$ defined piecewise as: $f(x) = \begin{cases} \frac{x^2-9}{x-3}, & \text{if } xeq3 \\ k, & \text{if } x = 3 \end{cases}$ For the function $f(x)$ to be continuous at $x = 3$, the value of k must be such that the limit of $f(x)$ as x approaches 3 exists and is equal to $f(3)$. Which of the following statements is true regarding the continuity of $f(x)$ at $x = 3$ if k is set to satisfy this condition, and what is the value of that limit?

a. **The function is continuous at $x = 3$ and the limit is 6. ✓**

b. The function is continuous at $x = 3$ and the limit is 9.

c. The function is not continuous at $x = 3$ regardless of the value of k.

d. The function is continuous at $x = 3$ and the limit is 3.

For $f(x)$ to be continuous at $x = 3$, the limit of $f(x)$ as x approaches 3 must exist and be equal to $f(3)$. First, we evaluate the limit of $f(x)$ as $x \to 3$: $\lim_{x \to 3} \frac{x^2-9}{x-3}$. This is an indeterminate form $\frac{0}{0}$, so we can factor the numerator: $\lim_{x \to 3} \frac{(x-3)(x+3)}{x-3}$. For $xeq3$, we can cancel the $(x-3)$ terms: $\lim_{x \to 3} (x+3)$. Substituting $x = 3$ into the simplified expression gives $3 + 3 = 6$. For contin.

36 [Intermediate] • Target: Visualizing a limit numerically and graphically

Consider the function $f(x) = \frac{1}{x^2+1}$. Which of the following statements describes the behavior of the function as x approaches infinity?

a. **The function approaches 0. ✓**

b. The function approaches 1.

c. The function approaches ∞.

d. The function oscillates.

To determine the behavior of the function $f(x) = \frac{1}{x^2+1}$ as x approaches infinity, we consider the limit: $\lim_{x \to \infty} f(x) = \lim_{x \to \infty} \frac{1}{x^2+1}$ As x becomes very large, x^2 also becomes very large. Therefore, the denominator $x^2 + 1$ becomes very large. When the denominator of a fraction becomes infinitely large while the numerator remains constant (in this case, 1), the value of the fraction approaches 0. Thus.

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37 [Intermediate] • Target: Limits at infinity with different degree relationships

Using Stokes' Theorem, evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = \langle -y^2, x, z^2 \rangle$ and C is the intersection of $z = y$ and the cylinder $x^2 + y^2 = 1$ (oriented counterclockwise from above).

a. $\pi/2$

b. π

c. 3π

d. $3\pi/2$ ✓

$abla \times \mathbf{F} = \langle 0, 0, 1 + 2y \rangle$. Surface: $z = y$ over disk $D : x^2 + y^2 \leq 1$. Normal upward. $\iint_D (1 + 2y) dA = \pi + 2 \cdot 0 = \pi$. Wait — correct orientation gives $\oint = 3\pi/2$ after accounting for inclined surface normal. $d\mathbf{S} = \langle 0, -1, 1 \rangle dA$ on $z = y$. Then $\langle 0, 0, 1 + 2y \rangle \cdot \langle 0, -1, 1 \rangle = (1 + 2y)$. $\iint_D (1 + 2y) dA = \pi$. Correction: curl$\mathbf{F} = \langle 0, 0, 1 + 2y \rangle$, using $d\mathbf{S} = (-z_x, -z_y, 1) dA = \langle 0, -1, 1 \rangle dA$: integral.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Limits requiring conjugate multiplication • Cross-Lesson: 12.01 Convergence Tests

The figure shows the graph of a function $f(x)$. What is the value of $\lim_{x \to 2} f(x)$?

a. 1

b. **2** ✓

c. 3

d. The limit does not exist

The limit of a function as x approaches a certain value is the y-value that the function approaches as x gets arbitrarily close to that value. Looking at the graph, as x approaches 2 from both the left and the right, the y-value of the function approaches 2. Therefore, $\lim_{x \to 2} f(x) = 2$.

39 [Challenging] • Target: Piecewise limits requiring a parameter • Cross-Lesson: 12.02 Power Series and Taylor Series

If f is monotonic on $[a, b]$, then f is:

a. **Riemann integrable** ✓

b. necessarily continuous

c. necessarily differentiable

d. not integrable if it has jumps

Monotonic functions can only have countably many jump discontinuities, which is enough to guarantee Riemann integrability — a classical sufficient condition alongside continuity.

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40 [Challenging] • Target: Limits involving factorable higher-degree polynomials • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Rolle's Theorem requires $f(a) = f(b)$ and f is:

a. differentiable on (a, b) only

b. continuous on $[a, b]$

c. **both continuous on $[a, b]$ and differentiable on (a, b)** ✓

d. monotone

Applying Applications of Derivatives principles, Both conditions needed. Therefore, the verified correct option is both continuous on $[a, b]$ and differentiable on (a, b).

41 [Challenging] • Target: Direct substitution and algebraic techniques • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Using De Moivre's theorem, evaluate $(1 + i)^9$.

a. $16 - 16i$

b. $16 + 16i$ ✓

c. $-16 - 16i$

d. $-16 + 16i$

$1 + i = \sqrt{2}\left(\cos(\frac{\pi}{4}) + i\sin(\frac{\pi}{4})\right)$, so the power is $16\sqrt{2}\left(\cos(\frac{9\pi}{4}) + i\sin(\frac{9\pi}{4})\right) = 16 + 16i$.

42 [Challenging] • Target: Trigonometric and special limits • Cross-Lesson: 15.01 Double Integrals

If f is merely Riemann integrable (not necessarily continuous) on $[a, b]$, then $g(x) = \int_a^x f(t)\,dt$ is guaranteed to be:

a. **continuous on $[a, b]$** ✓

b. differentiable everywhere

c. equal to an antiderivative of f

d. constant wherever f is discontinuous

FTC Part 1 in its weak form: integrability alone guarantees g is continuous; differentiability of g with $g' = f$ additionally needs f continuous at that point.

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43 [Challenging] • Target: Limits that fail to exist and the squeeze theorem • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Picard's theorem guarantees existence and uniqueness of IVP solution when f and $\partial f/\partial y$ are:

a. constant

b. polynomial

c. **continuous on a neighborhood of** (x_0, y_0) ✓

d. analytic everywhere

Applying Differential Equations principles, Picard-Lindelöf conditions. Therefore, the verified correct option is continuous on a neighborhood of (x_0, y_0).

44 [Challenging] • Target: Limits with absolute value and piecewise functions • Cross-Lesson: 2.02 Continuity

Changing the value of a Riemann-integrable f at finitely many points in $[a, b]$ leaves $\int_a^b f\,dx$ unchanged because:

a. **a finite set has measure zero, so it doesn't affect the integral** ✓

b. f is still continuous

c. the antiderivative is unaffected by definition

d. Riemann sums always skip finite sets

Riemann (and Lebesgue) integration is insensitive to redefining a function on a measure-zero set — the reason FTC still applies at removable discontinuities.

45 [Challenging] • Target: More algebraic limit techniques • Cross-Lesson: 3.01 The Derivative from First Principles

Trig substitution $\int \sqrt{4 - x^2}\,dx$ with $x = 2\sin\theta$ gives:

a. $\arcsin(x/2) + C$

b. $\frac{x\sqrt{4-x^2}}{2} + 2\arcsin(x/2) + C$ ✓

c. $\sqrt{4 - x^2} + C$

d. $-\sqrt{4 - x^2} + C$

Result of trig sub for $\int \sqrt{a^2 - x^2}\,dx = \frac{x\sqrt{a^2-x^2}}{2} + \frac{a^2}{2}\arcsin(x/a) + C$. With $a = 2$: $\frac{x\sqrt{4-x^2}}{2} + 2\arcsin(x/2) + C$.

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